

# AuDitors

**Presenter:** Anita Tsai

**Website:** SPOT: <https://from.so/>

Research Website for University of Michigan (School of Information  
and Computer Science)

# Introduction

# Research site for UMSI + UM CSE

Research focuses on factors  
that make programming  
tools usable and designing  
& building new tools for  
programmers.

[spot](#) [Research](#) [Team](#) [News](#) [Writing](#)

Computer programming skills are becoming indispensable for a variety of professions. However, computer programming is often prohibitively difficult to learn or do correctly. Many programming languages and tools were designed around the needs and limitations of the computers executing code, which can lead to design features that are counterintuitive for the people who use those tools.

We are a research group at the [University of Michigan](#), primarily in the [School of Information and Computer Science](#). Our research focuses on understanding the factors that make programming tools usable and designing & building new tools for programmers.

## News [\[+\]](#)

**Oct '24:** Excited to welcome Dr. Maryam Arab as a new Postdoctoral Researcher!

**Jun '24:** [Ashley](#) will be presenting her paper on representing code at scale at Learning @ Scale in Atlanta! 📄 📺

**Jun '24:** Congratulations to Dr. Lei Zhang on his wonderful Ph.D. defense! 🎉

**Apr '24:** [Lei Zhang](#) will join [NIIIT Informatics](#) as an Assistant Professor after a one-term postdoc at Princeton. Congratulations, Lei! 🎉

## People [\[+\]](#)



**Steve Oney**  
Associate Professor  
Michigan SI & CSE



**Maryam Arab**  
Postdoctoral Scholar  
Michigan SI



**Ashley Zhang**  
Ph.D. Student  
Michigan SI



**Jiacheng Zhang**  
Ph.D. Student  
Michigan SI



**Xiangyu Zhou**  
Ph.D. Student  
Michigan CSE

## Recent Publications [\[+\]](#)

Why we should care about  
accessibility

# 1.3 Billion

People or 16% of the world's population, or 1 in 6 of US population experiences significant disability.

[Source: World Health Organization](#)

# Users of SPOT

- Students
- Faculty
- Researchers
- Collaborators
- industry partners

# Prospective Students

- some students may not be able to navigate application information
- they may miss opportunities because they cannot access key content (due to structure)

# Students

- students with visual impairments may not be able to read publications
- students who rely on screen readers cannot interpret images or diagrams without “good” alt text



# Faculty / Researchers

- Faculty and research collaborators: explore datasets/access papers:
- key information may be difficult to find or interpret
- poor structure can make complex research content harder to navigate

# 100%

Universities should be all inclusive environments

Scope

# Tools

- Tools used:
  - WAVE (Chrome Extension)
  - aXe (Chrome Extension)

# Pages

- Pages Tested:
  - Homepage
  - Research page
  - Team
  - News
  - Maryam Arab
- Why?
  - They represented the most popular pages or pages with examples of complex content

# Manual Audit

- Reviewed all pages using **keyboard navigation** to ensure equal access to users
- Reviewed all pages at **200% magnification** to identify issues for low vision users
- Checked **color contrast for text on top of images** for users with color blindness & low vision
- Checked for actionable link text on pages

# What We Found

# Automated Test Result

## Errors:

### 5 aXe errors:

- Alt text shouldn't be the same as text
  - Sol: change the caption/text or alt text depending on which is more substantive
- Color blind people: can't see when you hover over
  - Sol: enlarge or add borderlines when hovered over

The screenshot shows a web browser window with the URL `https://from.so/`. The page content includes a navigation bar with links like 'spot', 'Research', 'Team', 'News', and 'Writing'. The main content area features a 'News' section with several articles, including one about Dr. Maryam Arab and another about Dr. Lei Zhang. Below the news section is a 'People' section with five profile pictures and names: Steve Onye, Maryam Arab, Ashley Zhang, Jiacheng Zhang, and Xiangyu Zhou. The browser's DevTools panel is open on the right, showing the 'aXe' tab. The 'Overview' tab displays a summary of accessibility issues, with a large '5' indicating the total number of issues. The 'Guided Tests' tab is also visible, showing a list of issues and their severity levels. The 'Total Issues: 5' is displayed at the bottom of the panel.

from.so

School Clubs Projects Internship

Notion ChatGPT Gmail Canvas GitHub WebDev

spot Research Team News Writing

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Ph.D. Student  
Michigan SI

Xiangyu Zhou  
Ph.D. Student  
Michigan CSE

Recent Publications [+]

DevTools  
axe-core 4.7.1.1

Sign up / Sign in

start new scan

Overview Guided Tests

Test Name Save Test

Test URL Re-run scan

https://from.so/

TOTAL ISSUES

Automatic Issues 5

Guided Issues 0

Manual Issues 0

Best Practice 5

Critical 0 Serious 0

Moderate 0 Minor 5

Best Practices: ON WCAG 2.1 AA

Total Issues: 5

Alternative text of images should not be repeated as text 5



# Automated Test Result

Errors:

0 WAVE errors

The screenshot displays the WAVE (Web Accessibility Evaluation Tool) interface. The top header shows the WAVE logo, "powered by WebAIM", and a language selector set to "en". Below the header, a status bar indicates "Styles: OFF" and "ON". The main content area is titled "Details" and shows a summary of results: 0 Errors, 0 Contrast Errors, 36 Alerts, 18 Features, 48 Structure, and 4 ARIA. A message states: "Congratulations! No errors were detected! Manual testing is still necessary to ensure compliance and optimal accessibility." Below this, the "AIM Score" is 9.4 out of 10. A list of alerts is shown, including "2 Redundant alternative text" and "1 Suspicious link text". The right side of the image shows a portion of a web page with a navigation menu (spot, Research, Team, News, Writing) and a "News" section with several articles, each featuring a "Headshot" of a person.

The following apply to the entire page:

spot Research Team News Writing

News [+]

Oct '24: Excited to welcome Dr. Maryam Arab as a new Postdoctoral Researcher!

"Headshot of Maryam Arab"

Jun '24: Ashley will be presenting her paper on representing code at scale at Learning @ Scale in Atlanta!

"Headshot of Ashley Zhang"

Jun '24: Congratulations to Dr. Lei Zhang on his wonderful Ph.D. defense!

"Headshot of Lei Zhang"

Code

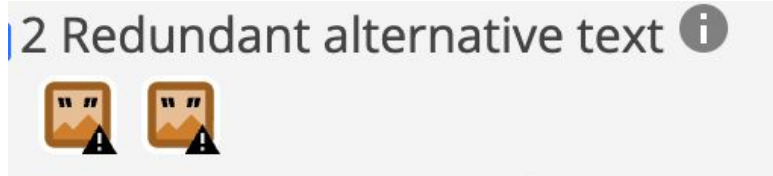
# Issue: Alt Text

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**Problem:** All the images have alt texts, but alternative text for an image is the same as nearby or adjacent text.

**Why This Is a Problem:** Alternative text that is the same as nearby or adjacent text will be presented multiple times to screen readers or when images are unavailable.

**Suggested Improvement:** Change either the alternative text or the adjacent text to eliminate the redundancy.



```
<a href="https://cse.engin.umich.edu/" target="_blank" rel="noopener">  
Michigan CSE
```

```
<br>  

```

# Issue: Publications Page Formatting (Large block of text + acronyms)

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**Problem:** The publications section appears as a long block of text instead of a list.

**Why This Is a Problem:** A long block of text is harder to read, and might be difficult to access for users with dyslexia or ADHD.

**Suggested Improvement:** Format publications using lists.

## **Publications:**

[Towards Inclusive Source Code Readability Based on the Preferences of Programmers with Visual Impairments](#)  
[Accessibility of UI Frameworks and Libraries for Programmers with Visual Impairments](#)  
[Understanding Accessibility and Collaboration in Programming for People with Visual Impairments](#)  
[Explore, Create, Annotate: Designing Digital Drawing Tools with Visually Impaired People](#)  
[Arboretum and Arbility: Improving Web Accessibility Through a Shared Browsing Architecture](#)

## **Publications:**

[FlowMatic: An Immersive Authoring Tool for Creating Interactive Scenes in Virtual Reality](#)  
[Improving Crowd-Supported GUI Testing with Structural Guidance](#)  
[Implementing Multi-Touch Gestures with Touch Groups and Cross Events](#)  
[Espresso: Building Responsive Interfaces with Keyframes](#)  
[InterState: A Language and Environment for Expressing Interface Behavior](#)  
[ConstraintJS: Programming Interactive Behaviors for the Web by Integrating Constraints and States](#)  
[FireCrystal: Understanding Interactive Behaviors in Dynamic Web Pages](#)

## **Publications:**

[Callisto: Capturing the "Why" by Connecting Conversations with Computational Narratives](#)  
[How Data Scientists Use Computational Notebooks for Real-Time Collaboration](#)

# Issue: Mobile Layout Issues

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**Problem:** On mobile devices, the page becomes difficult to read and navigate.

**Why This Is a Problem:** The page is very text heavy, and when zoomed in, content only appears on the left side.

**Suggested Improvement:** Adjust layout for mobile screens, so that content uses the full width of the screen.

spot Research Team News Writing

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# Issue: Icon use on website

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**Problem:** Icons are used on the website.

**Why This Is a Problem:** Icons can be problematic for accessibility, because their meaning may not be clear to all users and they may not be properly interpreted by assistive technologies such as screen readers.

**Suggested Improvement:** Provide text labels or accessible descriptions so users can clearly understand the meaning and function of the icons.

Connect with me



**Thank You!**